

Geometric Interpretation

If $\vec{F}$ is a velocity field,

flux = amount of matter through S per unit time

Dec 28: Divergence Theorem

Flux of $\vec{F}$ through surface S is defined by

$$\iint_S \vec{F} \cdot \hat{n} dS$$

If S is the graph of

$$z = f(x, y),$$

then

$$\hat{n} dS = \pm \langle -f_x, -f_y, 1 \rangle dx dy$$

Also,

$$\vec{u} \times \vec{v} = \Delta S \hat{n}$$

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Also,

$$\vec{u} \times \vec{v} = \Delta S \hat{n}$$

$$\vec{u} : (x, y, f(x, y))$$

$$\vec{u} \approx \langle \Delta x, 0, f_x \Delta x \rangle = \langle 1, 0, f_x \rangle \Delta x$$

$$\vec{v} \approx \langle 0, \Delta y, f_y \Delta y \rangle = \langle 0, 1, f_y \rangle \Delta y$$

$$\hat{n} \Delta S = \vec{u} \times \vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix}$$

$$\Delta x \Delta y \langle -f_x, -f_y, 1 \rangle$$

with $\hat{n}$ upwards.

Example

$$\vec{F} = z\mathbf{k}$$

through portion of paraboloid

$$z = x^2 + y^2$$

above unit disk, normal upwards.
(should be ≥ 0)

$$\begin{aligned} & \iint \vec{F} \cdot \hat{n} \, dS \\ &= \iint \langle 0, 0, z \rangle \cdot \langle -2x, -2y, 1 \rangle \, dx \, dy \\ &= \iint_S z \, dx \, dy \\ &= \iint_S (x^2 + y^2) \, dx \, dy \\ &= \iint (x^2 + y^2) \, dx \, dy = \int_0^{2\pi} \int_0^1 r^2 \cdot r \, dr \, d\theta \end{aligned}$$

More generally, given a parametric description of S ,

$$x = x(u, v)$$

$$y = y(u, v)$$

$$z = z(u, v)$$

$$\vec{r} = \vec{r}(u, v)$$

$$\hat{n} dS = du dv$$

Sides:

$$\frac{\partial \vec{r}}{\partial u} \Delta u \quad \text{and} \quad \frac{\partial \vec{r}}{\partial v} \Delta v$$

$$\begin{aligned} \hat{n} \Delta S &= \left(\frac{\partial \vec{r}}{\partial u} \Delta u \right) \times \left(\frac{\partial \vec{r}}{\partial v} \Delta v \right) \\ &= \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) \Delta u \Delta v \\ \hat{n} dS &= \pm \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) \Delta u \Delta v \end{aligned}$$

If we know a normal vector $\vec{N}$ (not necessarily unit) to the surface S .

Surface Element

$$\Delta A = \Delta S \cos \alpha$$

$$\cos \alpha = \frac{\vec{N} \cdot \vec{k}}{\|\vec{N}\| \|\vec{k}\|}$$

So,

$$\Delta S = \frac{1}{\cos \alpha} \Delta A = \frac{\|\vec{N}\|}{\vec{N} \cdot \vec{k}} \Delta A$$

$$\hat{n} \Delta S = \frac{\vec{N}}{\vec{N} \cdot \vec{k}} \Delta A$$

Hence,

$$\hat{n} dS = \pm \frac{\vec{N}}{\vec{N} \cdot \vec{k}} dx dy$$

Examples

$$z - f(x, y) = 0$$

Let

$$g(x, y, z) = z - f(x, y)$$

Then

$$\vec{N} = \nabla g = \langle -f_x, -f_y, 1 \rangle$$

Therefore,

$$\frac{\vec{N}}{\vec{N} \cdot \vec{k}} dx dy = \langle -f_x, -f_y, 1 \rangle dx dy$$

Divergence Theorem (Gauss' Theorem)

3D analogue of Green's theorem for flux.

If S is a closed surface enclosing a region D , oriented with $\hat{n}$ outward, and $\vec{F}$ is defined and differentiable everywhere in D , then

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_D \operatorname{div}(\vec{F}) dV$$

where

$$\operatorname{div}(P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) = P_x + Q_y + R_z$$

Example

Last time: flux of $z\mathbf{k}$ through S .

$$\iint_S z\mathbf{k} \cdot \hat{n} dS = \iiint_D \operatorname{div}(z\mathbf{k}) dV$$

$$0 + 0 + 1 = 1$$

$$\iiint_D dV = \operatorname{Vol}(D)$$

Lecture 29: Divergence Theorem

Flux of $\vec{F}$ through surface S is

$$\iint_S \vec{F} \cdot \hat{n} \, dS$$

If S is the graph of

$$z = f(x, y),$$

then

$$\hat{n} \, dS = \pm \langle -f_x, -f_y, 1 \rangle \, dx \, dy$$

$$\vec{u} \times \vec{v} = \Delta S \cdot \hat{n}$$

$\vec{u}$ from

$$(x, y, f(x, y))$$

to

$$(x + \Delta x, y, f(x + \Delta x, y))$$

$$\approx f(x, y) + \Delta x \, f_x$$

$$\vec{v} \approx \langle 0, \Delta y, f_y \Delta y \rangle = \langle 0, 1, f_y \rangle \Delta y$$

$$\hat{n} \Delta S = \vec{u} \times \vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix}$$

Example

$$\vec{F} = z\mathbf{k}$$

through portion of paraboloid

$$z = x^2 + y^2$$

above the unit disk.

$$\begin{aligned} & \iint_S \vec{F} \cdot \hat{n} \, dS \\ &= \iint \langle 0, 0, z \rangle \cdot \langle -2x, -2y, 1 \rangle \, dx \, dy \end{aligned}$$

$$\begin{aligned}
&= \iint_S z \, dx \, dy \\
&= \iint_S (x^2 + y^2) \, dx \, dy \\
&= \int_0^{2\pi} \int_0^1 r^2 \cdot r \, dr \, d\theta \\
&= \frac{\pi}{2}
\end{aligned}$$

More generally,
given a parametric description of S .

$$x = x(u, v)$$

$$y = y(u, v)$$

$$z = z(u, v)$$

$$\vec{r} = \vec{r}(u, v)$$

$$\hat{n} \, dS = du \, dv$$

Sides:

$$\frac{\partial \vec{r}}{\partial u} \Delta u = \left\langle \frac{\partial x}{\partial u}, \dots \right\rangle$$

and

$$\frac{\partial \vec{r}}{\partial v} \Delta v$$

$$\begin{aligned}
\hat{n} \, \Delta S &= \left(\frac{\partial \vec{r}}{\partial u} \Delta u \right) \times \left(\frac{\partial \vec{r}}{\partial v} \Delta v \right) \\
&= \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) \Delta u \Delta v
\end{aligned}$$

Therefore,

$$\hat{n} \, dS = \pm \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) du \, dv$$

If we know a normal $\vec{N}$ (not necessarily unit) to the surface S .

Plane

$$ax + by + cz = d$$

$$\vec{N} = \langle a, b, c \rangle$$

S given by equation.

Surface Element

$$\Delta A = \Delta S \cos \alpha$$

$$\cos \alpha = \frac{\vec{N} \cdot \vec{k}}{\|\vec{N}\| \|\vec{k}\|}$$

So,

$$\Delta S = \frac{1}{\cos \alpha} \Delta A = \frac{\|\vec{N}\|}{\vec{N} \cdot \vec{k}} \Delta A$$

$$\hat{n} \Delta S = \frac{\|\vec{N}\| \hat{n}}{\vec{N} \cdot \vec{k}} \Delta A = \pm \frac{\vec{N}}{\vec{N} \cdot \vec{k}} \Delta A$$

Therefore,

$$\hat{n} dS = \pm \frac{\vec{N}}{\vec{N} \cdot \vec{k}} dx dy$$

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$$\frac{\vec{N}}{\vec{N} \cdot \vec{k}} dx dy = \langle -f_x, -f_y, 1 \rangle dx dy$$

Divergence Theorem

3D analogue of Green's theorem.

If S is a closed surface enclosing a region D , oriented with $\hat{n}$ outward, and $\vec{F}$ defined and differentiable, then

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_D \operatorname{div} \vec{F} dV$$

where

$$\operatorname{div}(P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) = P_x + Q_y + R_z$$

Example

Flux of $z\mathbf{k}$ through sphere is

$$\begin{aligned} \frac{4\pi a^3}{3} \\ \iint_S z\mathbf{k} \cdot \hat{n} dS &= \iiint_D \operatorname{div}(z\mathbf{k}) dV \\ 0 + 0 + 1 &= 1 \\ &= \iiint_D dV = \operatorname{Vol}(D) \end{aligned}$$

Lecture 30: Line Integral in Space Curve

Diffusion Equation

u = concentration of substance

$\vec{F}$ = flow

Physical law:

$$\vec{F} = -k\nabla u$$

$$\frac{\partial u}{\partial t} = -\operatorname{div} \vec{F} = +k \operatorname{div}(\nabla u) = k\nabla^2 u$$

Therefore,

$$\frac{\partial u}{\partial t} = k\nabla^2 u$$

Diffusion equation.

Line Integrals in Space

Vector field

$$\vec{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$$

Curve C in space.

$$d\vec{r} = \langle dx, dy, dz \rangle$$

Work:

$$\int_C \vec{F} \cdot d\vec{r}$$

Therefore,

$$\int_C \vec{F} \cdot d\vec{r} = \int_C P dx + Q dy + R dz$$

To evaluate, parametrize and express

$$x, \quad y, \quad z$$

Example

$$\vec{F} = \langle yz, xz, xy \rangle$$

$$C: \quad x = t^3, \quad y = t^2, \quad z = t, \quad 0 \leq t \leq 1$$

$$dx = 3t^2 dt, \quad dy = 2t dt, \quad dz = dt$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_C yz dx + xz dy + xy dz$$

$$= \int_C t^3 \cdot 3t^2 dt + t^4 \cdot 2t dt + t^8 dt$$

$$= \int_0^1 6t^5 dt$$

$$= [t^6]_0^1 = 1$$

Example 2: Same $\vec{F}$, C :

$$\int_C = \int_{C_1} + \int_{C_2} + \int_{C_3}$$

For C_1, C_2 in the xy -plane,

$$z = 0, \quad dz = 0$$

$$\Rightarrow \int yz \, dx + xz \, dy + xy \, dz = 0$$

For C_3 ,

$$x = 1, \quad y = 1$$

$$dx = 0, \quad dy = 0$$

from 0 to 1

$$\int_{C_3} \vec{F} \cdot d\vec{r} = \int_{C_3} xy \, dz = \int_0^1 dz$$

In fact $\vec{F}$ is conservative,

$$\int_C \vec{F} \cdot d\vec{r} = \int_{C'} \vec{F} \cdot d\vec{r}$$

$$\vec{F} = \nabla(xyz)$$

$$= \langle yz, xz, xy \rangle$$

Knowing this, the Fundamental Theorem of Calculus for line integrals gives

$$\int_C \nabla f \cdot d\vec{r} = f(P_1) - f(P_0)$$

where

$$f(1, 1, 1) - f(1, 0, 1)$$